

CAMFS Unified Architecture (Method 2 + Patch)

LaTeX Notation, Mathematical Specification, Metrics & FedAMM

Comparative Reference

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Abstract

This reference document provides the complete mathematical notation, loss function formulations, aggregation equations, ground-truth label mappings [$c \in \{0 : \text{BG}, 1 : \text{NCR/NET}, 2 : \text{ED}, 4 : \text{ET}\}$], evaluation metrics (WT, TC, ET), policy matrix definitions, cold-start lineage rules, and an exhaustive comparative architectural analysis between **CAMFS** and **FedAMM (MICCAI)**.

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1 Notation & Symbol Definitions

Table 1: CAMFS Symbol and Variable Notation

Symbol	Meaning / Description	Type / Dimension
$\mathcal{M}$	Full modality universe {T1, T1ce, T2, FLAIR}	Set of symbols ($ \mathcal{M} = 4$)
$C = \{1, \dots, N\}$	Set of participating hospitals (clients)	Set of integers
$O(i) \subseteq \mathcal{M}$	Owned modality subset for client i	Subset of $\mathcal{M}$
$\text{Send}(i) \subseteq O(i)$	Modalities client i consents to transmit	Subset of $O(i)$
$\text{Recv}(i) \subseteq O(i)$	Modalities client i permits in received models	Subset of $O(i)$
S, S'	Fusion track identifier (keyed by modality subset)	Subset of $\mathcal{M}$
c	Ground-truth dataset class ID [$c \in \{0 : \text{BG}, 1 : \text{NCR}, 2 : \text{ED}, 4 : \text{ET}\}$]	Discrete class index
C_{cls}	Total segmentation classes $\{0 : \text{BG}, 1 : \text{NCR}/\text{NET}, 2 : \text{ED}, 4 : \text{ET}\}$	Scalar ($C_{\text{cls}} = 4$)
Encoder_m	Unimodal feature encoder for modality m	Neural net parameters θ_{E_m}
Encoder_m^*	Frozen, static post-Phase-1 encoder snapshot	Static weights (requires_grad=False)
Proto_m^c	Global prototype for modality m and class $c \in \{0, 1, 2, 4\}$	Vector $\mathbb{R}^{256}$
$\text{Proto}_{m,i}^{c,t}$	Local prototype for modality m , class c from client i	Vector $\mathbb{R}^{256}$
$\text{FusionHead}_{S'}$	Track- S' cross-attention feature fusion module	Neural net parameters $\theta_{F_{S'}}$
$\text{Decoder}_{S'}$	Track- S' segmentation decoder module	Neural net parameters $\theta_{D_{S'}}$
$\text{FusedProto}_{S'}^c$	Track- S' fused bottleneck prototype for class c	Vector $\mathbb{R}^{256}$
$R_{\text{send}}(i, m)$	Modality encoder upload policy gate	Binary $\{0, 1\}$
$R_{\text{recv}}(i, j, m)$	Modality encoder download policy gate	Binary $\{0, 1\}$
$R_{\text{send}}^{\text{track}}(i, S')$	Track fusion upload policy gate	Binary $\{0, 1\}$
$R_{\text{recv}}^{\text{track}}(i, j, S')$	Track fusion download policy gate	Binary $\{0, 1\}$
$R_{\text{contribute}}(i, S)$	Superset client opt-in gate for non-native track S	Binary $\{0, 1\}$
$z_m(p)$	Bottleneck feature embedding at spatial location p	Vector $\mathbb{R}^{256}$
P_m^c	Bottleneck positions with ground-truth class $c \in \{0, 1, 2, 4\}$	Set of 2D coordinates
τ	Contrastive temperature hyperparameter	Scalar ($\tau = 0.1$)
λ_1, λ_2	Loss weighting parameters	Scalars ($\lambda_1 = 0.5, \lambda_2 = 0.5$)
$\text{Lineage}(\theta)$	Cumulative set of modalities touching parameter θ	Subset of $\mathcal{M}$
$\text{detach}(\cdot)$	Graph-level stop-gradient operator	Tensor operation

2 Formulas & Equations

2.1 Phase 1: Unimodal Encoder Alignment

For client i , owned modality $m \in O(i)$, and spatial position p with ground-truth class $c(p) \in \{0 : \text{BG}, 1 : \text{NCR}, 2 : \text{ED}, 4 : \text{ET}\}$, spatial feature embeddings $z_m(p)$ are aligned with class prototypes Proto_m^c via location-wise InfoNCE loss:

$$\mathcal{L}_{\text{unimodal-align}}(z_m) = -\frac{1}{|P_m|} \sum_{p \in P_m} \log \frac{\exp(\text{sim}(z_m(p), \text{Proto}_m^{c(p)})/\tau)}{\sum_{c' \in \{0,1,2,4\}} \exp(\text{sim}(z_m(p), \text{Proto}_m^{c'})/\tau)} \quad (1)$$

where $\text{sim}(u, v) = \frac{u^T v}{\|u\|_2 \|v\|_2}$ denotes cosine similarity.

Local prototype estimates for class $c \in \{0, 1, 2, 4\}$ are computed as spatial centroid means:

$$\text{Proto}_{m,i}^{c,t} = \frac{1}{|P_{m,i}^c|} \sum_{p \in P_{m,i}^c} z_m(p) \quad (2)$$

Federated prototype aggregation for client k is evidence-weighted by pixel count $|P_{m,j}^c|$:

$$\text{Proto}_m^{c,t+1} \rightarrow k = \sum_{j: m \in \text{Send}(j), R_{\text{recv}}(k,j,m)=1} \frac{|P_{m,j}^c|}{N_m^{c,(k)}} \text{Proto}_{m,j}^{c,t} \quad (3)$$

2.2 Phase 1 Stopping Criterion & Freeze Procedure

Phase 1 completes after T_1 minimum rounds when global prototype drift across all classes $c \in \{0, 1, 2, 4\}$ falls below threshold ϵ for K consecutive rounds:

$$\frac{1}{4C_{\text{cls}}} \sum_{m \in \mathcal{M}} \sum_{c \in \{0,1,2,4\}} \|\text{Proto}_m^{c,t} - \text{Proto}_m^{c,t-1}\|_2 < \epsilon \quad \text{for } K \text{ consecutive rounds} \quad (4)$$

At transition, encoder weights are locked ($\theta_{E_m}^*$) and computation graph detachment is enforced:

$$z_m = \text{detach}(\text{Encoder}_m^*(x_m)) \implies \frac{\partial z_m}{\partial \theta_{E_m}^*} \equiv 0 \quad (5)$$

2.3 Phase 2: Track-Isolated Fusion Training

Phase 2 computes task segmentation loss (Dice + Cross-Entropy) and fused prototype alignment for ground-truth classes $c \in \{0, 1, 2, 4\}$:

$$\mathcal{L}_{\text{task}}(y, \hat{y}) = \mathcal{L}_{\text{CE}}(y, \hat{y}) + \mathcal{L}_{\text{Dice}}(y, \hat{y}) \quad (6)$$

$$\mathcal{L}_{\text{fusion-align}}(z_{S'}) = -\frac{1}{|P_{S'}|} \sum_{p \in P_{S'}} \log \frac{\exp(\text{sim}(z_{S'}(p), \text{FusedProto}_{S'}^{c(p)})/\tau)}{\sum_{c' \in \{0,1,2,4\}} \exp(\text{sim}(z_{S'}(p), \text{FusedProto}_{S'}^{c'})/\tau)} \quad (7)$$

2.4 Send-Gated Track Aggregation Equations

Track fusion updates are aggregated strictly over clients whose declared **Send policy** matches track identity S' :

$$\theta_{\text{FusionHead}_{S'}}^{t+1} \rightarrow k = \sum_{\substack{j: \text{Send}(j)=S', \\ R_{\text{send}}^{\text{track}}(j,S')=1, \\ R_{\text{recv}}^{\text{track}}(k,j,S')=1}} \frac{n_j}{N_{S'}^{(k)}} \theta_{\text{FusionHead}_{S',j}}^t \quad (8)$$

3 Consolidated Loss Functions

3.1 Phase 1 Overall Client Loss

$$\mathcal{L}_i^{\text{Phase 1}} = \lambda_1 \sum_{m \in O(i)} \mathcal{L}_{\text{unimodal-align}}(z_m) \quad (9)$$

3.2 Phase 2 Overall Client Loss

$$\mathcal{L}_i^{\text{Phase 2}} = \mathcal{L}_{\text{task}}(y_i, \hat{y}_i) + \lambda_2 \mathcal{L}_{\text{fusion-align}}(z_{S'}) \quad (10)$$

Note: λ_1 's term is strictly absent in Phase 2 because encoder parameters $\theta_{E_m}^*$ are frozen (`requires_grad = False`) and graph-detached.

4 Governance, Policy & Cold-Start Tables

Table 2: Policy Matrix Definitions and Operational Semantics

Policy Matrix	Governance Level	Operational Semantics
$R_{\text{send}}(i, m)$	Modality Encoder Upload	1 if client i transmits modality m encoder updates.
$R_{\text{recv}}(i, j, m)$	Modality Encoder Download	1 if client i accepts modality m updates originating from j .
$R_{\text{send}}^{\text{track}}(i, S')$	Track Fusion Upload	1 if client i transmits track S' fusion/decoder deltas.
$R_{\text{recv}}^{\text{track}}(i, j, S')$	Track Fusion Download	1 if client i accepts track S' updates originating from j .
$R_{\text{contribute}}(i, S)$	Multi-Track Superset Opt-in	1 if client i ($O(i) \supset S$) runs masked passes for track S .

Table 3: Final Recommended 4-Hospital Experimental Setup

Client	Owned $O(i)$	Send $\text{Send}(i)$	Receive $\text{Recv}(i)$	Tested Mechanism
Hospital 1	{T1, T1ce, T2, FLAIR}	{T1, T1ce, T2, FLAIR}	{T1, T1ce, T2, FLAIR}	Full Track S_1 & $R_{\text{contribute}}$ for
Hospital 2	{T1, T1ce, T2}	{T1, T2}	{T1, T1ce, T2}	Send-Gated Re-keying to S'_2 &
Hospital 3	{T1, FLAIR}	{T1, FLAIR}	{T1, FLAIR}	Receive Filtering & Purity Pro
Hospital 4	{T1, T1ce, T2}	{T1, T1ce, T2}	{T1, T1ce, T2}	Dynamic Phase 2 Join & Tier

Table 4: 4-Tier Cold-Start Initialization Policy & Lineage Rule

Tier	Name	Initialization Procedure & Compliance Condition
Tier 1	Exact Match	Pull active track S' parameters directly if $\text{Send}(i_{\text{new}}) = S'$.
Tier 1b	Composable Stitching	Slice channels from nearest superset track if $\text{Lineage}(\theta_{\text{seed}}) \subseteq \text{Send}(i_{\text{new}})$.
Tier 2	Warm-Start	Fine-tune sliced parameters for W local rounds on frozen encoders.
Tier 3	Fresh Initialization	Random init of fusion head on frozen encoders if Lineage Rule fails.

5 Ground-Truth Labels, Clinical Sub-Regions & Metrics

5.1 Ground-Truth Label Mapping

BraTS NIfTI masks contain 4 ground-truth label classes $c \in \{0, 1, 2, 4\}$:

- **Class $c = 0$ (BG):** Background and healthy brain tissue.
- **Class $c = 1$ (NCR/NET):** Necrotic and non-enhancing tumor core.
- **Class $c = 2$ (ED):** Peritumoral edema.
- **Class $c = 4$ (ET):** Enhancing tumor (Label 3 is unused).

Table 5: BraTS Clinical Evaluation Sub-Region Mapping

Clinical Region	Symbol	Dataset Label Composition
Whole Tumor	WT	Classes $c \in \{1, 2, 4\}$ (NCR/NET + ED + ET)
Tumor Core	TC	Classes $c \in \{1, 4\}$ (NCR/NET + ET)
Enhancing Tumor	ET	Class $c = 4$ only (ET)

5.2 Evaluation Metrics Formulations

5.2.1 Dice Similarity Coefficient (DSC)

For ground-truth binary mask Y and predicted mask $\hat{Y}$ over region $R \in \{\text{WT}, \text{TC}, \text{ET}\}$:

$$\text{DSC}_R(Y, \hat{Y}) = \frac{2|Y_R \cap \hat{Y}_R|}{|Y_R| + |\hat{Y}_R|} \times 100\% \quad (11)$$

5.2.2 Compliance-Performance Gap ($\Delta_{\text{compliance}}$)

Measures the explicit accuracy cost of hard consent compliance relative to an unrestricted oracle:

$$\Delta_{\text{compliance}}^{(i)} = \text{DSC}_{\text{Oracle}}^{(i)} - \text{DSC}_{\text{CAMFS}}^{(i)} \quad (12)$$

5.2.3 Purity Probe Classification Accuracy

A binary classifier f_{probe} is trained on bottleneck embeddings z_m or fused features $z_{S'}$ to predict exposure to disallowed modality m^* :

$$\text{Acc}_{\text{probe}} = \mathbb{P}[f_{\text{probe}}(z) == \text{is_exposed}(m^*)] \quad (13)$$

Compliance invariant: $\text{Acc}_{\text{probe}} \equiv \text{Null Baseline (Chance Level)}$.

6 Comparative Analysis: CAMFS vs. FedAMM (MICCAI)

Table 6: Exhaustive Architectural Comparison: CAMFS vs. FedAMM

Dimension	FedAMM (MICCAI)	CAMFS Unified (Method 2 + Patch)
Core Objective	Performance optimization under missing modalities.	Verifiable compliance under institutional consent constraints.
Prototype Definition	Modality-combination prototypes (tied to client clusters).	Unimodal Anatomical Prototypes: Prototypes represent 1 modality $m \in \mathcal{M}$ and 1 ground-truth class $c \in \{0 : \text{BG}, 1 : \text{NCR}, 2 : \text{ED}, 4 : \text{ET}\}$ ($\text{Proto}_m^c \in \mathbb{R}^{256}$).
Training Lifecycle	Single-phase joint training (encoders + fusion co-adapted).	Two-phase sequential training (unimodal alignment $\rightarrow$ freeze $\rightarrow$ isolated fusion).
Gradient Leakage	Soft Leakage: Task loss backprops into shared encoders; un-consented signals bleed across clients.	Zero Leakage: Encoders frozen and ‘detach()’ed before fusion; zero gradient backprop.
Routing Mechanism	Learned weighted aggregation by modality proportion.	Exogenous policy matrices ($R_{\text{send}}, R_{\text{recv}}, \text{Send}(i), \text{Recv}(i)$).
Restricted Send	Not supported (assumes Own = Send).	Supported via Send-Gated Track Re-keying to $S' + \text{Private Local Head}$.
Governance Artifact	None (opaque aggregation weights).	Inspectable Provenance Ledger & Lineage Rule.
Evaluation Metrics	Dice (WT, TC, ET).	Dice (WT, TC, ET) + Compliance-Performance Gap + Purity Probe.

7 Assumptions & Scope Conditions

Table 7: Methodological Assumptions and Scope Limitations

#	Label	Description / Scope Boundary
1	Honest Execution (A1)	Clients faithfully follow declared policy matrices. Byzantine/malicious attacks are out of scope.
2	Frozen Lineage Safety	Post-Phase-1 encoder weights $\theta_{E_m}^*$ are static and cannot leak cross-modal signal in Phase 2.
3	2D Processing	Slice-based 2D U-Net backbone default for fast convergence and lower memory overhead.
4	BraTS Modalities	Evaluated on BraTS MRI sequences (T1, T1ce, T2, FLAIR) as pseudo-modalities.